How-To Library

Informatica

Enabling SAML authentication with Okta SSO for web applications

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Abstract

You can enable users to log into Informatica web applications using single sign-on. This article explains how to configure single sign-on in an Informatica domain using Security Assertion Markup Language (SAML) v2.0 and the Okta SSO identity provider.

Supported Versions

- Informatica Data Engineering Integration 10.5
- Informatica Enterprise Data Catalog 10.5
- Informatica Enterprise Data Preparation 10.5
- Informatica Metadata Manager 10.5

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Overview

You can configure Security Assertion Markup Language (SAML) authentication for the Informatica domain using the Okta SSO identity provider.

An identity provider is an entity that provides authentication as a consumable service by applications. Platforms like Amazon Web Services (AWS) and Microsoft Azure support various third-party identity providers to authenticate requests by applications on their platforms.

SAML is an XML-based data format for exchanging authentication information between a service provider and an identity provider. In an Informatica domain, an Informatica web application is the service provider.

You can configure the following Informatica web applications to use SAML authentication:

- Informatica Administrator
- Informatica Analyst
- Metadata Manager

- Enterprise Data Catalog
- Enterprise Data Preparation

Informatica 10.5 supports Okta SSO version 2021.02.3.

Note: SAML authentication cannot be used in an Informatica domain configured to use Kerberos authentication.

Prerequisites

Verify the following prerequisites before you begin integrating Okta SSO with Informatica.

Verify that your installation of Okta SSO is version 2021.02.3 or higher.

Get the Okta dashboard URL that corresponds to your Okta user account. For example: `https://<user identifier>-admin.okta.com/admin/dashboard`

Deploy Informatica in the same network as the Okta SSO deployment. Identify a machine where the Informatica domain is deployed and note the host name and IP address.

Configure Okta SSO

Perform the steps in this section to integrate Okta SSO with Informatica.

Download and install the Active Directory agent

Download and install the Active Directory agent to enable application monitoring.

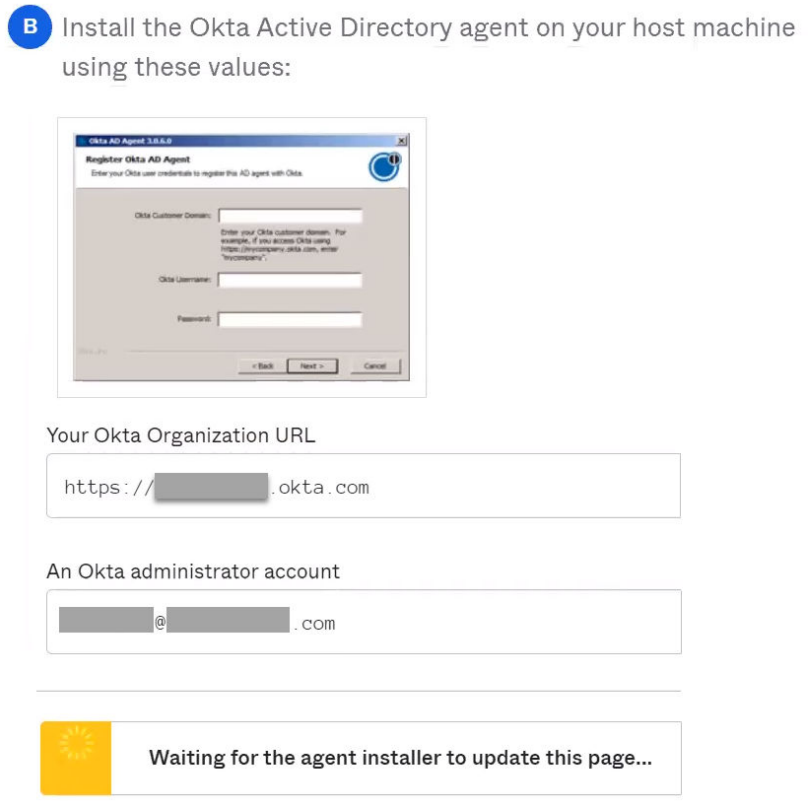
1. In the Okta dashboard, browse to **Directory** > **Directory integrations**.
2. Click **Add Directory** > **Add Active Directory**.
3. Click **Set Up Active Directory**.
4. Download and install the Active Directory agent to a directory on the Virtual Machine (VM) that runs the Okta application agent:
   a. Click **Download Agent**.

      Okta downloads an .exe file to your computer.

   b. Run the downloaded .exe file.

   c. Follow the instructions under "B. Install the Okta Active Directory agent on your host machine using these values."

      Notice that values for the **Register Okta AD Agent** dialog box appear beneath the screenshot of the dialog box. The following image shows the instructions:



In the **Register Okta AD Agent** dialog box, configure the following properties and then click **Next**.

Property	Description
Okta Customer Domain	Enter the value displayed under "Your Okta Organization URL."
Okta Username	Enter the value displayed under "An Okta administrator account."
Password	Enter the password for the Okta administrator account.

The agent is installed and registered.

Import users from Active Directory

Verify the agent and then import Active Directory users to Okta.

1. Check the agent status.
   If the status is Active, proceed to import users.
2. Import users from Active Directory. See the Okta documentation.

Create an application

Create a SAML application in the Okta portal.

1. Log in to the Okta SSO administration console dashboard at the following URL: `https://<Okta host>/admin/dashboard`.

2. Browse to **Applications** > **Admin**.
3. Click **Create new application**.
4. Select **SAML 2.0** and click **Next**.
5. Configure the following properties for the application:

Property	Description
Appname	Enter a name for the application.
Single Sign-On URL	Enter the URL for the Informatica Administrator tool.
Requestable URLs: Index, Recipient, Destination	Enter the URL for the Informatica Administrator tool. If you want to enable additional Informatica domains to use Okta SAML SSO, click **Add Another** and enter the Informatica Administrator URL for each domain.
Audience URI	Identifier to use to represent the Informatica service provider. Okta uses the same value for the Service Provider (SP) Entity URI property. Enter a string to identify Informatica as the service provider. For example: `Infa`.
Default Relay State	The URL that users will be directed to after a successful authentication through SAML. Do not provide a value for this property.
Name ID format	The username format that Okta uses in the SAML response. Accept the default value `Unspecified`.
Application username	Determines the default value for a user's application username. Okta uses this as the user name in the authentication assertion statement. Select **AD SAM account name**.
Response	Enables Okta to sign authentication request responses from a service provider. **Note:** Informatica does not support response signing for the Okta SSO identity provider.
Assertion signature	Choose to require assertion signing. When you enable assertion signing, Okta signs assertions of authenticity using a symmetric key. Enabling this feature is recommended.
Assertion encryption	Optionally choose to require encrypted assertions. When you enable encrypted assertions, Okta signs assertions of authenticity using a symmetric key.
Enable Single Logout	Enables end users to use a single action to sign out of the Okta session and a configured application. Verify the property is not enabled. Default is not enabled.
Authentication context class	The method that Okta uses to authenticate the user with a specific authentication mechanism. Identifies the SAML authentication context class for the assertion's authentication statement. Select **PasswordProtectedTransport**.
Honor Force Authentication	Whether or not to require users to authenticate through the web app. Select **Yes**. Default is No.
SAML Issuer ID	Accept the default value: `http://www.okta.com/${org.externalKey}`

6. Configure at least one attribute statement. The attribute statement is included in SAML assertions from Otka.

Property	Description
Name	Username of an account imported from Active Directory. When you enter the literal string `username`, Okta takes the user account name from the user that is assigned to the application in the **Assignments** tab.
Name Format	Choose Unspecified.
Value	Value for the attribute. Enter the following string: `String.substringBefore(user.login, "@")` The string takes the Active Directory (AD) account user name and filters the @ symbol and anything that follows it. For example, if the Active Directory user is firstlastname@PLATFORMKRB.COM, the value of the **Name** attribute is `firstlastname`.

7. Click **Create app**.
8. Click **View Setup Instructions** to get the Okta SSO URL for the application. Copy this URL to use later when you configure the Informatica domain. Example:
`https://<Okta host>/app/<application name>/<unique key>/sso/saml`
9. Click **Next**.
10. On the "Help Okta Support" page, select the statement that begins "I'm an Okta customer..." The remaining questions are optional.
11. Scroll to the bottom and click **Finish**.

Assign users to the application

Assign the application that you created to an imported Active Directory user.

1. From the **Application** page, click the **Assignments** tab.
2. Click **Assign** > **Assign to people**.
The wizard displays the users that you imported from Active Directory.
3. Select a user to assign to the application. Optionally, use the Search control to find the user that you want to assign to the application.

4. Click **Assign** for each user to assign.
   Enter an alias for each user in the **Assign <application name> to People** dialog box and click **Save and Go Back**.

   Repeat this step for each user.

   The following image shows the **Assign <application name> to People** dialog box:



The Assignments tab displays the list of users assigned to the application.

Download the truststore certificate

Download the truststore certificate from the Okta implementation for import to the Informatica domain.

1. In the Okta dashboard, browse to **Admin** > **Applications** > **Applications** and select the application you created.
2. Click the **Sign On** tab.
   The Okta portal opens to a page titled **How to Configure SAML 2.0 for <application name>**.

   The page displays the following properties, with the value for each already populated:

   Identity Provider Single Sign-On URL. URL that the service provider uses to connect with Okta.

   Identity Provider Issuer. URL that Okta uses to furnish the truststore certificate when an app requests it.

   X.509 Certificate. Encoded text of the truststore certificate.
3. Click **Download certificate** and save the .jks file to a temporary directory.

Enable Okta SSO for the Informatica domain

Configure elements on the domain to enable integration with Okta SSO.

Import the truststore certificate

The okta.cert file validates the SAML assertion signature. Import the file to the Informatica domain truststore.

Locate the okta.cert file in the location where you downloaded it, and then import it to the domain. Use the keytool utility to import the file. For information about keytool, see
Import the Certificate into the Truststore Used for SAML Authentication.

1. Locate the okta.cert file that you downloaded previously. See "Download the truststore certificate" on page 7.
2. Use keytool to import the certificate file to the domain.
3. Restart the domain.

The certificate is imported to the truststore in the following path: `<Informatica home directory>/source/services/shared/security/`

For information about using keytool to import the truststore certificate, see the *Informatica Security Guide*.

Update the domain SAML configuration

Use the infasetup updateDomainSAMLConfig command to enable SAML authentication on the domain and specify the identity provider URL.

For example: `infasetup updateDomainSamlConfig -saml true -iu https://<Okta host>:<port>/idp/SSO.saml2`

Note: The example does not contain all required options. See the information about infasetup updateDomainSAMLConfig in the *Informatica Command Reference*. For example, the -spid option is required for Informatica releases 10.2.2 and higher. The option specifies the relying party trust name.

Optional: Configure the certificate alias on gateway nodes

If you use the assertion signing feature, update gateway nodes with the Okta truststore certificate alias.

The certificate alias refers to the truststore certificate that enables signed assertion.

1. Run the following command to configure the certificate alias:
   `./infasetup.sh updateGatewayNode -saml true -asca <alias>`
2. After the command runs, restart the Informatica domain.

Perform additional domain configuration tasks

Use infacmd to update the domain namespace and synchronize security domains.

Enable LDAP connectivity to synchronize Active Directory users

Use the addLDAPConnectivity command to synchronize Active Directory users.

The following example shows the command to synchronize Active Directory users.

```
infacmd.sh addLDAPConnectivity -dn ISPDomain -un Administrator -pd Administrator -sdn Native -la
<domain host>:389 -lp <user ID> -lc "<user name>" -lt MicrosoftActiveDirectory -ms 5000 -lcn LDAP2
```

Add the namespace for Active Directory users

Use the addNamespace command to add a namespace, creating a security domain in the Informatica domain.

A namespace on the Informatica domain contains a set of users imported from a user repository. When you create a namespace, the Informatica Administrator tool can access all the Active Directory users that you imported. This enables single sign-on for the Administrator tool.

The following example shows the command to add a namespace:

```
infacmd.sh addNameSpace -dn ISPDomain -un Administrator -pd Administrator -SecurityDomain Native -
usb cn=users,dc=platformkrb,dc=com -uf objectclass=user -gsb "cn=users,dc=platformkrb,dc=com" -gf
"objectCategory=group" -hp <domain host>:<port> -ns LDAP_NameSpace1 -lcn LDAP2
```

Synchronize security domains

Use the syncSecurityDomains command to synchronize the domain with the users that you imported from Active Directory.

The following example shows the command to synchronzie security domains:

```
infacmd.sh SyncSecurityDomains -dn ISPDomain -un Administrator -pd Administrator -sdn Native -sn
LDAP_NameSpace1
```

Assign the administrator role to a user

Use the assignRoleToUser command to assign the administrator role to an existing LDAP user.

The following example shows the command to assign the administrator role:

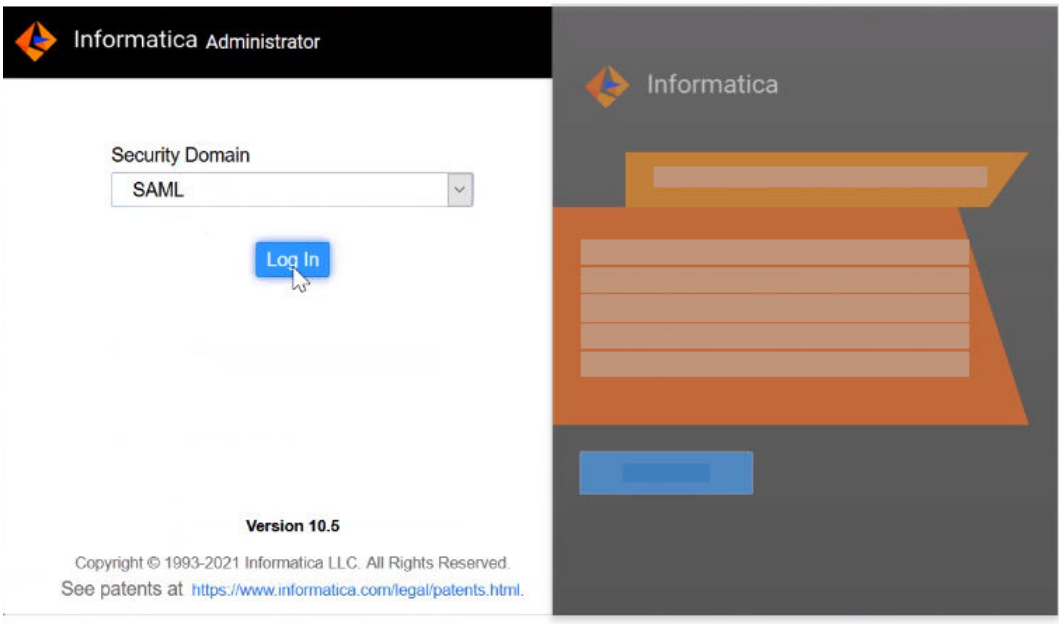
```
infacmd.sh assignRoleToUser -dn ISPDomain -un ispadmin <user name> -sdn Native -eu <user id> -esd LDAP <namespace name> -rn Administrator -sn ISPDomain
```

Log In to the Administrator tool

Now you can log into the Informatica domain using Okta SSO.

1. In a browser, go to the Administrator tool URL.
2. Log in using the security domain that you created for Okta SSO.

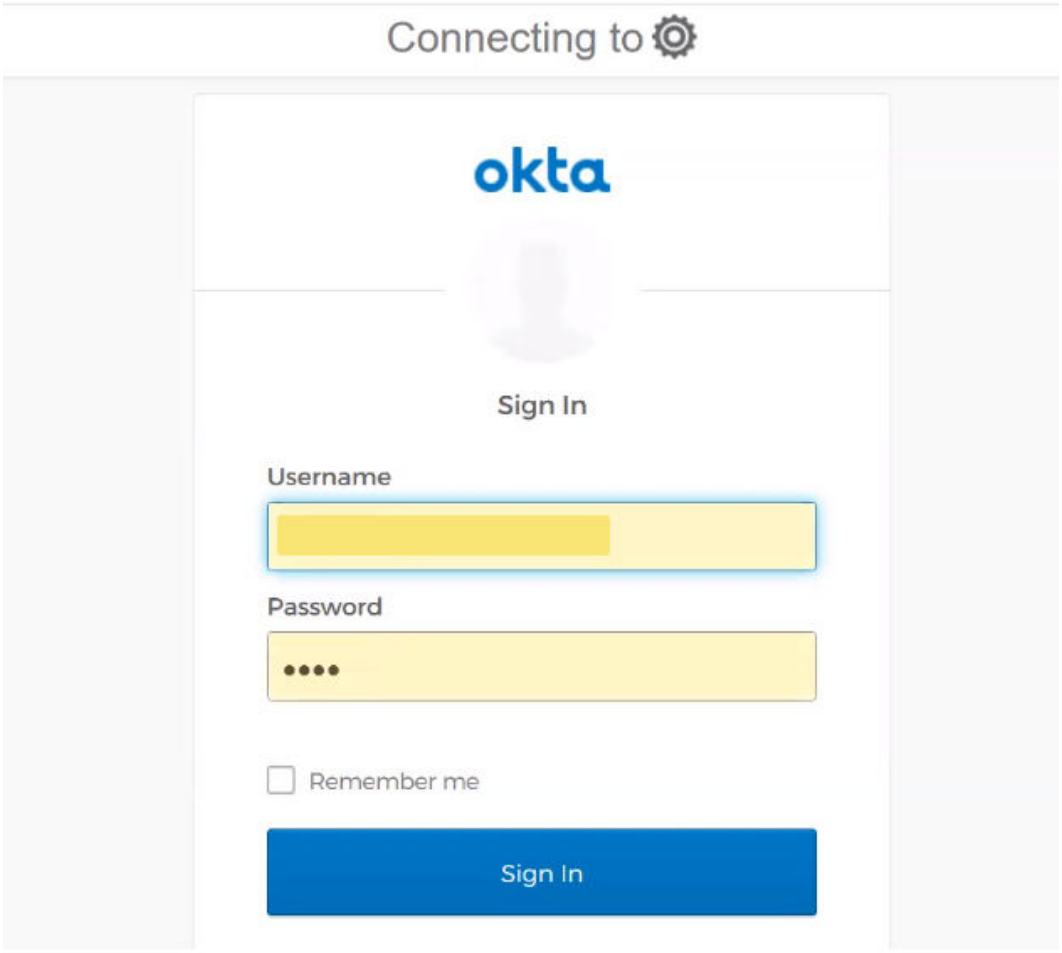
   For example, you created a security domain for Okta SSO named "SAML." The following image shows the Administrator tool login screen:



3. Click **Log In**.

   The browser redirects you to the Okta SSO login page.

4. Log in to Okta using the administrator login that you designated with the AssignRoleToUser command. The following image shows the Okta login screen:



The Administrator tool opens.

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